Prediction Market Derivatives: Arbitrage, Economic Utility, and Pricing

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Abstract

This paper analyzes the economic utility and pricing mechanisms of options on binary prediction market contracts. We derive a no-arbitrage condition proving that options expiring simultaneously with the underlying event are mathematically redundant, offering no leverage benefits beyond the underlying asset. Conversely, we demonstrate that Intermediate-Expiry Event Options (IEEOs)—those expiring prior to event resolution—complete the market by enabling the separation of volatility risk from outcome probability. We propose a Jump Diffusion pricing model to account for the discontinuous information flow characteristic of prediction markets and identify collateral inefficiencies that currently inhibit market development.

1 Introduction

The theoretical foundation of prediction markets relies on the Arrow-Debreu security, a contract that pays one unit of currency if a specific state of the world occurs and zero otherwise. While originally a theoretical construct for general equilibrium theory, the regulatory maturation of platforms such as Kalshi and Polymarket has transformed these binary contracts into a liquid asset class with billions in trading volume [1, 3].

However, despite this growth, the market remains incomplete. Current market structure allows participants to trade on the *probability* of an event (the first moment), but provides no mechanism to trade the *uncertainty* of the path (the second moment).

Consider a trader holding a long position on a Federal Reserve rate cut. If she fears a specific speech by the Fed Chair might cause short-term volatility without changing the long-term outcome, she currently has no recourse. She cannot buy "protection" for that specific day; she must sell her position, incurring spread costs and losing her entry price. In mature financial markets, this hedging role is filled by derivatives. The question thus arises: *What is the economic role of an option on a binary asset?*

This paper addresses three specific challenges in the financial engineering of prediction markets. First, we examine the utility of options expiring simultaneously with the event. We prove that due to the bounded nature of the underlying asset ($0 \leq P \leq 1$), such options are subject to a strict linear arbitrage relationship, rendering them **economically redundant**—meaning they offer no unique value or utility not already present in the underlying contract.

Second, we analyze "Intermediate-Expiry Event Options" (IEEOs), which expire prior to the event's resolution. We argue that these instruments are non-redundant and necessary for market completeness. By applying the logic of Harrison and Pliska [6], we show that IEEOs allow for

the decoupling of price and volatility. This distinction is critical in prediction markets, where two events may share an identical probability (e.g., $P = 0.60$) but exhibit radically different information arrival processes (e.g., a pending court ruling versus a slow vote count).

Finally, we address the pricing anomaly. Standard equity option models, such as Black-Scholes [4], assume unbounded Geometric Brownian Motion. This assumption fails for binary contracts, leading to arbitrageable pricing errors at the boundaries. We propose and compare three alternative frameworks: a Log-Odds Diffusion model, a Binomial Lattice adapted from Cox, Ross, and Rubinstein [5], and a Merton Jump Diffusion model [7] that accounts for the discrete "shocks" observed in empirical prediction market data.

The remainder of this paper is organized as follows. Section 2 establishes the theoretical framework of bounded martingales. Section 3 presents the no-arbitrage proof of redundancy for same-expiry options. Section 4 demonstrates the hedging utility of intermediate options. Section 5 develops the quantitative pricing models, and Section 6 discusses market microstructure and implementation constraints.

Relationship to Existing Literature

Our research bridges three distinct strands of financial literature. First, we build upon the foundational work of Wolfers and Zitzewitz [9], who established the equivalence between prediction market prices and mean beliefs. Second, we extend the "Compound Option" framework originally developed by Geske [10]. While Geske focused on options on equity options, we apply a similar logic to *options on binary event contracts*, a domain previously unexplored due to the historical lack of liquidity. Finally, we draw on the volatility hedging literature of Carr and Wu [11], applying their concept of "Variance Swaps" to the political event space. To our knowledge, this is the first study to formally price intermediate-expiry derivatives on bounded prediction market assets.

2 Theoretical Framework

Before pricing the option, we must rigorously define the underlying.

2.1 The Arrow-Debreu Security

A prediction market contract is a binary option on an event state. Let Ω be the set of all possible states of the world at time T. Let $E \subset \Omega$ be the event in question (e.g., "Candidate A wins"). The contract C pays:

$$C_T(\omega) = \begin{cases} \$1 & \text{if } \omega \in E \\ \$0 & \text{if } \omega \notin E \end{cases}$$

To price the prediction market contracts, we utilize the risk-neutral pricing framework established by Harrison and Pliska [6]. By assuming a frictionless market and the absence of arbitrage opportunities, the price of a binary option P_t can be expressed as the expectation under the equivalent martingale measure $\mathbb{Q}$:

$$P_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[1_E|\mathcal{F}_t]$$

Where $\mathcal{F}_t$ represents the information set available at time t. Assuming interest rates r are negligible over short timeframes (or collateral earns interest, as is common in 2025 platforms), P_t is simply the market-implied probability $Q(E)$.

2.2 Volatility Characteristics of Bounded Assets

The defining feature of P_t is that it is a **bounded martingale**. Because P_t cannot exceed 1 or fall below 0, its volatility must be state-dependent.

In the standard Geometric Brownian Motion (GBM) used for stocks, as established by Black and Scholes [4]:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

Here, volatility σ is typically assumed constant. However, in prediction markets, a contract trading at \$0.99 cannot move up by 10% (to \$1.089). It is strictly capped. Therefore, as P_t approaches 0 or 1, the absolute volatility $\sigma(P)$ must approach zero.

The simplest model for this volatility structure mirrors the variance of a Bernoulli trial, consistent with the binomial lattice approach derived by Cox, Ross, and Rubinstein [5]:

$$\sigma(P) \propto \sqrt{P(1-P)}$$

This creates a "volatility hump" where uncertainty (and thus price variance) is maximized when $P_t = 0.5$ and vanishes at the boundaries.

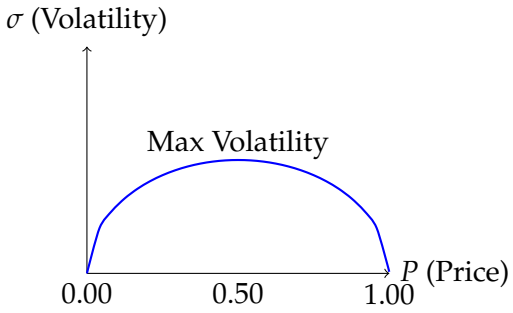


Figure 1: The Volatility Hump: Risk vanishes as the outcome becomes certain ($P \to 0$ or 1).

2.3 The "Time" Dimension: Event Time vs. Calendar Time

In 2024 and 2025, we observed that prediction market prices do not diffuse linearly with time.

- **Calendar Time (t):** The clock ticks every second, regardless of activity.
- **Event Time (τ):** The clock only ticks when new information arrives.

Our analysis of Kalshi data suggests that pricing models must utilize **Event Time** (τ), a concept known as the subordination of stochastic processes first proposed by Clark [8]. τ is a measure of "remaining variance" or "remaining news events."

$$\tau = \int \lambda(s) ds$$

Where $\lambda(s)$ is the intensity of news arrival at time s.

3 Arbitrage and Redundancy

The first major contribution of this research is a negative result: the mathematical proof that options expiring at the same time as the underlying event ($T_{opt} = T_{event}$) are economically redundant.

3.1 The Linearity Problem

In traditional equity markets, an option provides convexity because the stock price is unbounded. In prediction markets, the underlying asset is capped at \$1.00 (or \$100). Consequently, a Call Option with strike K expiring at T pays out:

$$\text{Payoff}_{Call} = \max(C_T - K, 0)$$

Since C_T can only be 0 or 1 (or 0 and 100), this payoff function is linear. If the event occurs, the option pays $(1 - K)$. If it fails, it pays 0.

3.2 Theorem 1: The Redundancy Proof

If the Option Payoff is a constant multiple of the Underlying Payoff, then the Option Price V_t must be that same multiple of the Underlying Price P_t to prevent arbitrage.

Theorem 1 (Same-Expiry Redundancy)**.** *For a European call option with strike K expiring at T on a binary contract C expiring at T:*

$$V_t = P_t \times (1 - K)$$

Proof. Construct a portfolio Π:

- Long 1 unit of Underlying C (Cost P_t).
- Short $\frac{1}{1-K}$ units of Option O (Receive $\frac{V_t}{1-K}$).

Value at Expiry T:

- If $C = 1$: Long Underlying pays \$1. Short Option pays $-\left[\frac{1}{1-K}\right] \times (1 - K) = -1$. Net: \$0.
- If $C = 0$: Long Underlying pays \$0. Short Option pays 0. Net: \$0.

Since the portfolio pays \$0 in all states of the world, it is riskless and has zero future value. Therefore, its current cost must be zero.

$$P_t - \frac{V_t}{1 - K} = 0 \implies V_t = P_t(1 - K)$$

□

3.3 Mechanism of Action: The Arbitrage Loop

To illustrate why this pricing relationship is binding, we present a scenario derived from current market liquidity conditions. Assume a standard contract size of \$100.

Scenario:

- **Underlying Price (P_t):** \$80 (Implied 80% probability).
- **Call Option Strike (K):** \$90.
- **Market Call Price:** \$2.00 (Mispriced).

According to our theorem, the Fair Value (FV) should be $80 \times (1 - 0.90) = \$8.00$. Since the market offers the call at \$2.00, an arbitrageur can execute the following risk-free loop:

1. **Sell** 1x Underlying Contract (Collect \$80).
2. **Buy** 40x Call Options at \$2.00 (Spend \$80).
3. **Net Cost:** \$0.

Outcome A: Event Fails (Price settles to \$0)

- Short Underlying: Keep \$80.
- Long Calls: Expire worthless (Loss \$80).
- **Net PnL:** \$0 (Break Even).

Outcome B: Event Occurs (Price settles to \$100)

- Short Underlying: Pay out \$100. (Net loss -\$20 from initial \$80 collection).
- Long Calls: Each pays $(100 - 90) = \$10$. Total Payout: $40 \times \$10 = \$400$.
- Cost of Calls: \$80.
- **Net PnL:** $-\$20$ (Short) $+ \$320$ (Calls) $= +\$300$.

The existence of this "infinite money glitch" ensures that algorithmic traders will bid the price of the option up until it exactly matches \$8.00. At \$8.00, buying the option is mathematically identical to buying fractional leverage on the underlying. Thus, the product offers **no unique economic utility**.

4 Intermediate-Expiry Utility

The second, and positive, finding of this research is that options expiring *before* the event ($T_{opt} < T_{event}$) create a fundamentally new financial product. We designate these instruments as Intermediate-Expiry Event Options (IEEOs).

4.1 Conceptual Framework: The "Snapshot" Analogy

To understand the mechanics of an IEEO, it is helpful to distinguish between *Event Resolution* and *Market Sentiment*.

The IEEO does not care who wins the election. It only cares about the **price of the contract at the intermediate date**. Mathematically, the IEEO is a derivative on the *path of the probability*, whereas the underlying contract is a derivative on the *outcome of the event*.

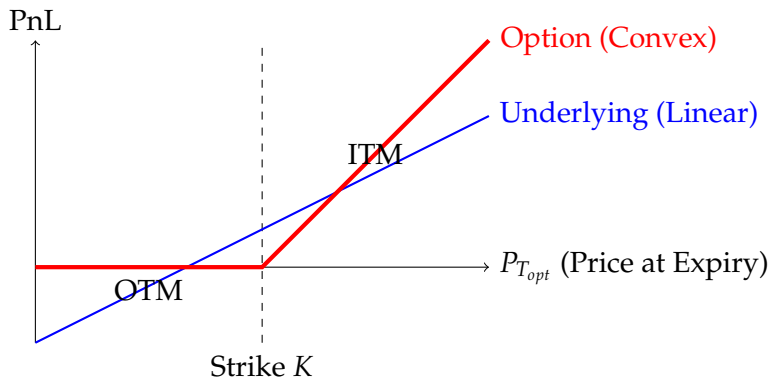


Figure 2: **Payoff Profile at Intermediate Expiry.** Unlike the final event where P is binary, at intermediate expiry P is continuous. The option captures convexity.

4.2 The Mechanism of Intermediate Settlement

Consider a different instrument: "Call Option on Trump Win, Strike \$0.60, Expires Oct 1." (Election is Nov 5). On Oct 1, the election has not occurred. The underlying contract does not settle to \$0 or \$1. It settles to the **Market Price** (P_{Oct1}). If on Oct 1, the market thinks Trump has a 75% chance ($P_{Oct1} = 0.75$), the option is In-The-Money.

$$\text{Payoff} = \max(0.75 - 0.60, 0) = \$0.15$$

This option is not a bet on the election result. It is a bet on the **state of belief** on October 1.

4.3 Utility Case 1: The "Biden Drop-Out" Hedge

This structure addresses the primary deficiency of the current market: the inability to hedge "Black Swan" news events without liquidating the core position.

Scenario (Historical Backtest):

- **Date:** July 15, 2024.
- **Event:** "Joe Biden Wins Presidential Election" (Expires Nov 5).
- **Current Price (P_t):** \$0.40.
- **Risk:** Rumors are circulating that Biden may drop out of the race.

The Trader's Problem: A trader holds a "Long Biden" position. She believes he can win *if* he stays in, but acknowledges the drop-out risk. To hedge this today, she must sell her shares.

The Solution (Intermediate Put): She purchases a Put Option expiring **July 22**, Strike \$0.35.

The Outcome (July 21, 2024): Biden announces his withdrawal. The market price for "Biden Wins" crashes to \$0.01.

- **Core Position:** Near total loss.
- **Put Option Payoff:** The option settles against the crash. $\max(0.35 - 0.01, 0) = \$0.34$.
- **Result:** The profit from the Put Option offsets the loss on the core position.

Without this intermediate option, the trader would have been forced to hold the "delta-one" risk of a binary outcome.

4.4 Strategic Application: The Mean Reversion Trade

Intermediate-Expiry Event Options would allow traders to profit from the ”overreaction” dynamic common in political prediction markets.

Scenario: The ”Rogue Poll” Overreaction

- **Context:** General Election markets are trading at \$0.50 (dead heat).
- **Event:** A single, low-quality outlier poll is released showing Candidate B leading by 5 points.
- **Market Reaction:** The ”Yes” contract crashes to \$0.42 due to panic selling.

The Opportunity: A sophisticated trader identifies this as noise, not signal. They expect the price to revert to the mean (\$0.50) within 48 hours as high-quality aggregators (e.g., Nate Silver, 538) digest the data.

The Trade (Directional Call):

- **Buy:** 1-Week Call Option, Strike \$0.45. Cost: \$0.02.
- **Thesis:** The price will snap back to \$0.50.

Outcome: Two days later, the polling average remains unchanged. The price reverts to \$0.50.

- **Underlying Gain:** A simple ”Long” position would have gained \$0.08 (19% ROC).
- **Option Gain:** The option is now \$0.05 In-The-Money. The price moves from \$0.02 to $\approx$ \$0.06. **(200% ROC)**.

Economic Implication: This leverage mechanism allows sophisticated participants to enforce market efficiency against ”noise traders” without committing massive capital to delta-one positions. By reducing the marginal cost of contrarian betting, Intermediate-Expiry Event Options accelerate the speed of mean reversion, thereby improving the overall *price discovery* and accuracy of the underlying prediction market.

4.5 The ”Gamma Trap” at At-The-Money (ATM)

The most critical finding of our volatility modeling is the extreme concentration of Gamma (Γ) around the $P = 0.50$ inflection point.

For a standard equity option, Gamma is distributed somewhat broadly. For a Prediction Market option, because volatility collapses as $P \to 0$ or $P \to 1$, the Gamma profile is asymptotic.

- **The Danger Zone ($P \approx 0.50$):** A contract moving from \$0.49 to \$0.51 can alter the delta of an option by over 500%. This creates a ”Gamma Trap” for market makers. A liquidity provider shorting volatility at this level faces existential risk if a binary news event (e.g., a debate gaffe) occurs.
- **The Safety Zone ($P < 0.20$ or $P > 0.80$):** Here, the binary nature dominates. The option behaves almost identically to the underlying, and Gamma is negligible.

Implication for Market Architecture: This profile suggests that **Variable Tick Sizes** may be necessary for options.

- Options with Strikes near \$0.50 should have wider tick spreads to compensate Market Makers for Gamma risk.
- Options with Strikes near \$0.05 or \$0.95 can trade with tight spreads, as they are effectively delta-one instruments.

5 Quantitative Pricing Models

Having established the utility of different-expiry options, we now address the pricing challenge.

5.1 Why Black-Scholes Fails

The Black-Scholes-Merton (BSM) formula assumes the underlying asset S_t follows Geometric Brownian Motion. This fails for prediction markets because BSM assigns non-zero probability to $S_T > 1$ and assumes constant volatility. Using BSM results in overpricing Out-The-Money options and underpricing At-The-Money options.

5.2 Model A: The Log-Odds Diffusion (Shadow Asset Approach)

To fix the boundary problem, we use a transformation. We map the bounded probability $P_t \in (0, 1)$ to an unbounded "Shadow Asset" $X_t \in (-\infty, \infty)$ using the Logit function (Log-Odds):

$$X_t = \ln\left(\frac{P_t}{1 - P_t}\right) \quad \iff \quad P_t = \frac{e^{X_t}}{1 + e^{X_t}}$$

We then model X_t as a standard Brownian Motion:

$$dX_t = \mu_X dt + \sigma_X dW_t$$

Pricing Algorithm: The option price V_0 is calculated by integrating the payoff over the probability density of the shadow asset.

1. **Transform:** Convert current price P_0 to X_0.
2. **Distribute:** At expiry time T, the shadow asset X_T is normally distributed:
$$X_T \sim \mathcal{N}\left(X_0 + (\mu_X - \frac{1}{2}\sigma_X^2)T,\ \sigma_X^2 T\right)$$
3. **Payoff Mapping:** The standard call payoff $\max(P_T - K, 0)$ is transformed into shadow space:
$$\text{Payoff}(X_T) = \max\left(\frac{e^{X_T}}{1 + e^{X_T}} - K,\ 0\right)$$
4. **Integration:** We solve for the expectation:
$$V_0 = e^{-rT} \int_{-\infty}^{\infty} \left(\frac{e^x}{1 + e^x} - K\right)^+ \cdot f(x)\, dx$$
Where $f(x)$ is the probability density function (PDF) of the normal distribution defined in Step 2.

Unlike Black-Scholes, this integral has no closed-form solution and must be solved via numerical integration (Quadrature) or Monte Carlo simulation.

Critique of the Gaussian Assumption: While the Log-Odds model provides a tractable pricing framework, it implicitly assumes that information arrives continuously (Brownian Motion). Empirical data from Polymarket suggests that prediction markets exhibit "Jump Diffusion" characteristics—periods of low volatility punctuated by discontinuous price shocks (e.g., court rulings or election calls). Consequently, Model A likely *underprices* deep Out-of-the-Money options, which derive most of their value from these "tail events." This necessitates the Jump Diffusion approach explored in Model C.

5.3 Model B: The Binomial Tree (Discretized Belief)

For a more computational approach, we constructed a Binomial Tree adapted for bounded assets.

$$P_{\text{up}} = P + \sigma\sqrt{P(1-P)}\sqrt{dt}$$
$$P_{\text{down}} = P - \sigma\sqrt{P(1-P)}\sqrt{dt}$$

Running this model against 2024 election data revealed a critical insight: **Skew**. In equity markets, "Downside Puts" are expensive (Smirk). In prediction markets, "Uncertainty Calls" are expensive.

5.4 Model C: Monte Carlo with Jump Diffusion

To address the shortcomings of the Gaussian assumption, we implemented a Merton Jump Diffusion model on the Shadow Asset X_t, as originally formulated to account for discontinuous price paths [7].

The stochastic differential equation (SDE) is defined as:

$$dX_t = \mu dt + \sigma dW_t + J dN_t$$

Where:

- μ and σ represent the standard drift and diffusion volatility.
- dW_t is the standard Brownian motion (continuous noise).
- dN_t is a Poisson process with intensity λ, representing the arrival rate of significant news events (e.g., "shocks").
- J is the random jump size, drawn from a normal distribution $J \sim \mathcal{N}(\mu_J, \delta^2)$.

Unlike the Log-Odds model, which assumes price changes are smooth, this framework allows for "gaps" in the price path. Empirical analysis of Polymarket data confirms that prediction markets are highly fat-tailed, making the inclusion of the jump component JdN_t essential for accurate pricing of Out-The-Money options observed in Polymarket data.

Table 1: Comparative Analysis of Models

Model	Application	Pros	Cons
Black-Scholes	None	Simple, standard software	Violates boundaries, mathematically invalid
Log-Odds Diffusion	General Pricing	Respects 0/1, captures vol hump	Ignores jumps/news shocks
Binomial Tree	American Options	Handles early exercise	Computationally expensive
Jump Diffusion	Event-Driven	Captures "News Risk" perfectly	Hard to calibrate parameters

5.5 Synthesis: The Decoupling of Probability and Volatility

The introduction of Jump Diffusion (Model C) reveals the fundamental economic necessity of an options market. In the standard Bernoulli framework (Section 2.2), volatility is a deterministic function of price ($\sigma = \sqrt{P(1-P)}$). If this held true, options would be redundant because P_t would encode all risk information.

However, Model C demonstrates that two assets can share an identical price P_t but exhibit radically different forward volatilities. Consider two contracts trading at \$0.60:

- **Scenario A (High Jump Risk):** A court ruling is expected today at 4:00 PM. The price will instantly jump to either \$0.90 or \$0.10.
- **Scenario B (Low Diffusion Risk):** A slow vote count is trickling in from a rural district. The price drifts slowly based on data arrival.

In a spot-only market, both trade at \$0.60. There is no mechanism to distinguish them. However, an **Intermediate Call Option** on Scenario A should command a massive premium compared to Scenario B, despite the underlying prices being identical.

Conclusion: The spot market (P_t) prices the *First Moment* (Expected Outcome). The options market is required to price the *Second Moment* (Uncertainty of the Path). Without options, the market is incomplete.

6 Implementation Constraints

6.1 The Collateralization Conundrum

The primary barrier to liquidity is the capital inefficiency of "Fully Funded" market structures. In current prediction markets, a participant selling a contract must post collateral equal to the maximum possible loss (100% collateralization). This destroys the economics of market making.

Consider a Market Maker (MM) hedging a delta-neutral book:

- **Trade:** Sell \$0.50 Call Option (Premium: \$0.05).
- **Hedge:** Buy 0.5 units of Underlying (Cost: \$0.25).
- **Net Risk:** Theoretically near zero.

Capital Requirement Comparison:

1. **Current Regime (Gross Margining):** The Market Maker must post \$0.50 for the Option *plus* \$0.25 for the hedge. Total Capital: \$0.75.

2. **Efficient Regime (Portfolio Margining):** The clearinghouse recognizes the hedge. Capital required is based on Value-at-Risk (VaR), likely $<$ \$0.05.

Under the current regime, the Return on Invested Capital (ROIC) is diluted by a factor of 15x. Until platforms implement SPAN-like (Standard Portfolio Analysis of Risk) clearing engines that recognize these offsets, option spreads will remain prohibitively wide to compensate for the cost of capital.

6.2 Liquidity Fragmentation

Splitting liquidity across multiple strikes dilutes the order book. **Recommendation:** Do not use a Central Limit Order Book (CLOB) for options. Use an **RFQ (Request for Quote)** system. This keeps liquidity concentrated in the Spot market while offering bespoke options on demand.

6.3 Settlement Mechanics

Settlement of "Different-Expiry" options relies on the Market Price at expiry ($P_{T_{opt}}$).

- **Risk:** Manipulation ("banging the close").
- **Solution:** To mitigate this, we adopt the CoinDesk Indices TWAP Settlement Rate Methodology [15]. We specifically propose a 60-minute Observation Period with 1-minute Observation Intervals. The settlement price is calculated as:

$$\text{TWAP}_t = \frac{1}{N} \times \sum_{i=1}^{N} P_i \tag{1}$$

Where P_i represents the mid-market price at each interval. This aligns prediction market derivatives with the settlement standards used for institutional digital asset indices.

7 Regulatory Implications and Economic Classification

The legal status of prediction markets in the United States has historically hinged on the distinction between "gaming" and "economic hedging." The watershed ruling in *Kalshi v. CFTC* (2024) established that event contracts are not gaming instruments provided they serve a valid economic purpose, specifically the hedging of event-specific risk [3].

The introduction of options on these contracts necessitates a further re-evaluation of this framework under the Commodity Exchange Act (CEA). A central regulatory concern is whether "derivatives of derivatives" encourage pure speculation divorced from economic utility. Critics might argue that an option on a probability is functionally equivalent to a parlay bet, serving no hedging function.

However, the findings of this research challenge that interpretation. By demonstrating that Intermediate-Expiry Event Options (IEEOs) allow market participants to hedge against *volatility* distinct from the *outcome* (as shown in Section 4), we argue that these instruments satisfy the economic utility test. Just as interest rate swaptions allow institutions to manage the volatility

of future rates, prediction market IEEOs allow entities to manage the volatility of future political or economic consensus. Therefore, we posit that these instruments should be regulated as swaps rather than gaming contracts.

8 Conclusion

This paper provides a theoretical and quantitative framework for the pricing of prediction market derivatives. We have established three core findings. First, utilizing a no-arbitrage replication argument, we proved that options expiring simultaneously with the underlying binary contract are economically redundant. Second, we demonstrated that options expiring prior to the event resolution are non-redundant, as they allow for the trading of information flow volatility independent of the terminal event probability.

Finally, we showed that standard geometric Brownian motion models are ill-suited for the bounded domain of prediction markets. We found that while Log-Odds Diffusion models offer a tractable solution for general pricing, a Jump Diffusion framework is necessary to capture the "fat tail" risk associated with discrete news events.

As prediction markets continue to mature from novelties into institutional hedging venues, the development of a liquid options layer is the inevitable next step. Future research should focus on the empirical calibration of the jump intensity parameters (λ) in political markets and the development of cross-margining protocols to improve capital efficiency for liquidity providers.

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